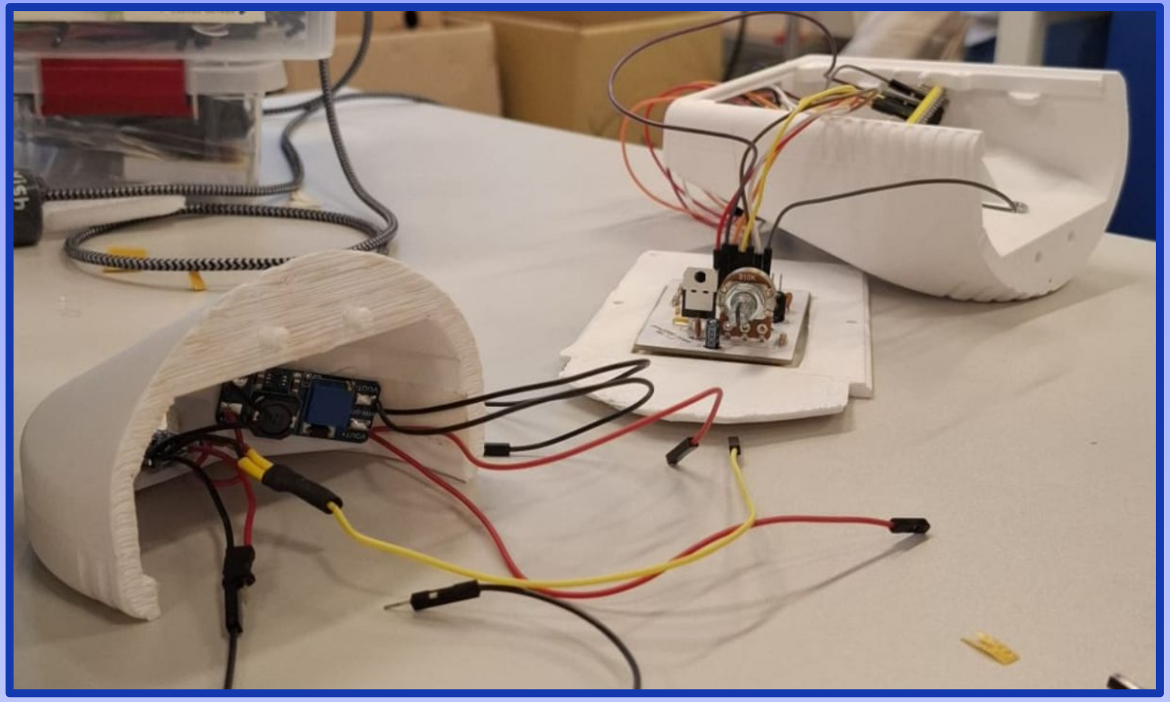
BTFL MOUSE



BLUETOOTH TOUCH FEEDBACK
LOW-POWER MOUSE

HARDWARE & SENSORS

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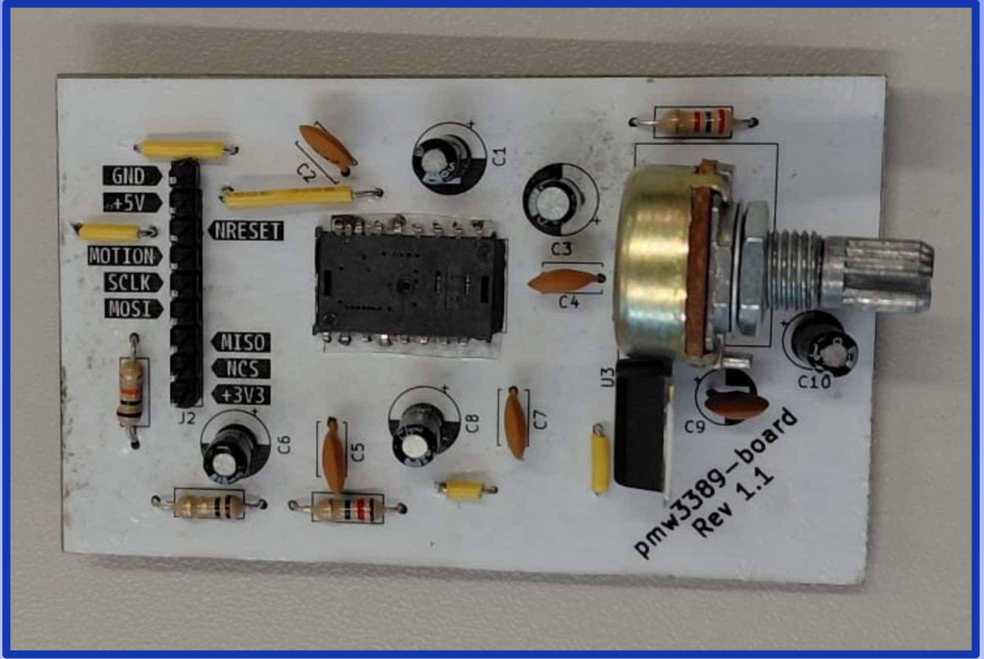
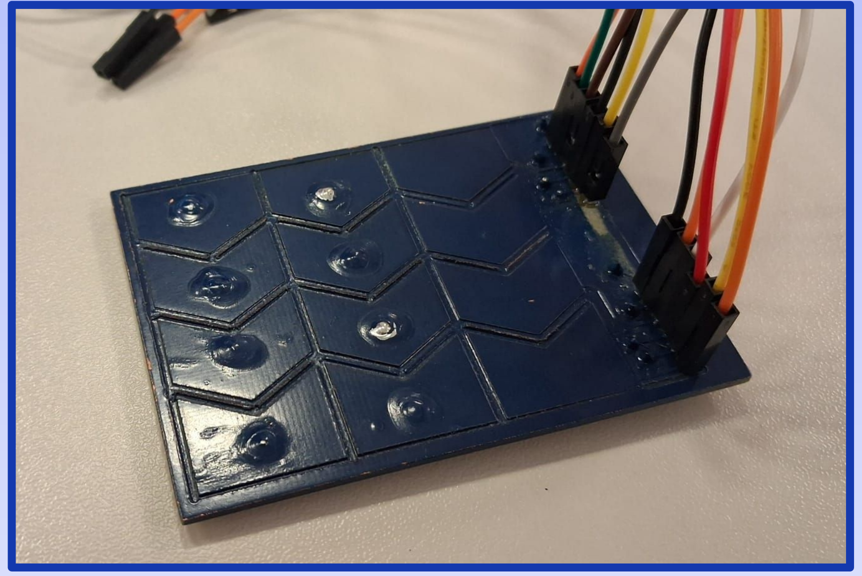


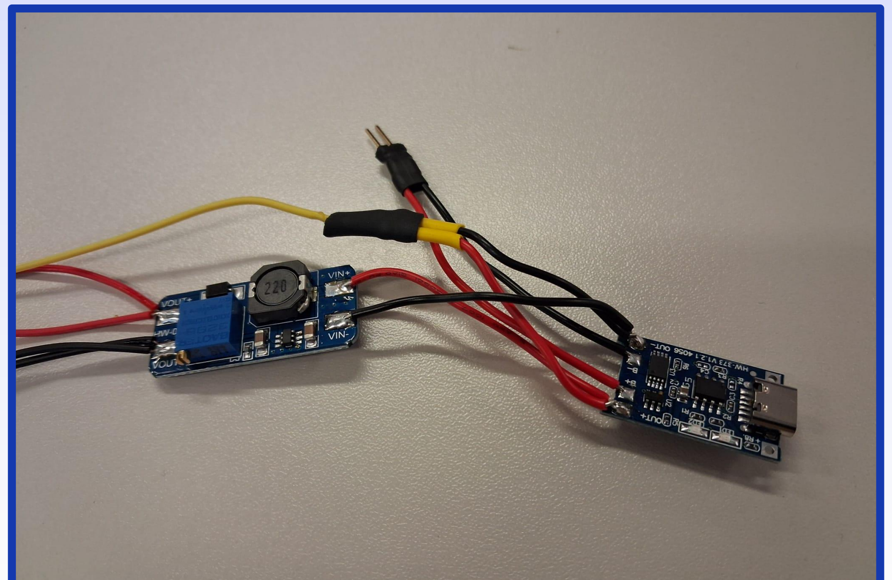
Optical Sensor

- PMW3389DM-T3QU
- CPI: 50-16,000
- Custom PCB



Capacitive Touch Surface

- 4x3 electrode matrix
- Custom PCB

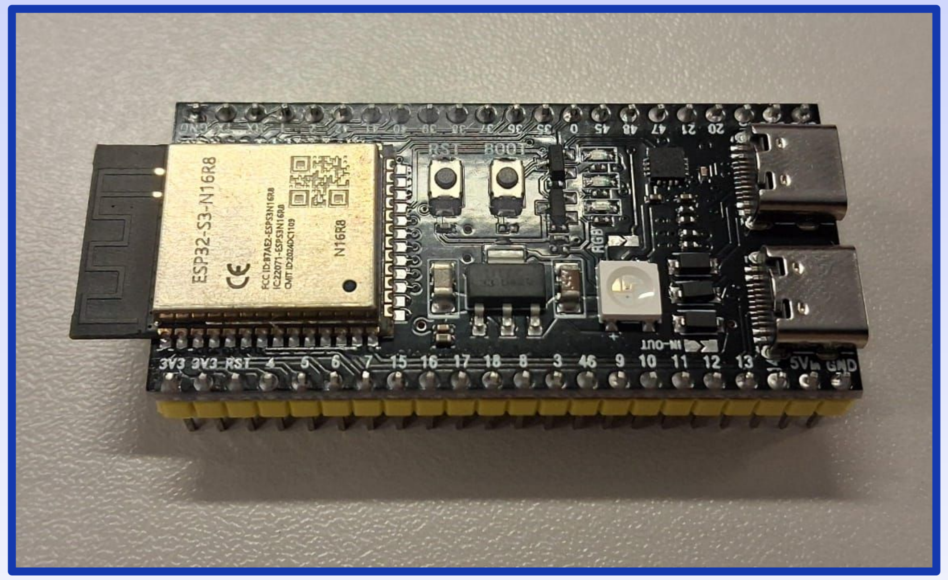


Power Management

- Lithium Battery
- Rechanging circuit
- Boost converter
- Battery level

Microcontroller

- ESP32-S3
- Dual-core Xtensa LX7
- 240 MHz max clock
- 8MB Flash, 2MB PSRAM
- WiFi/BT Antenna



WORKFLOW STATE MACHINE

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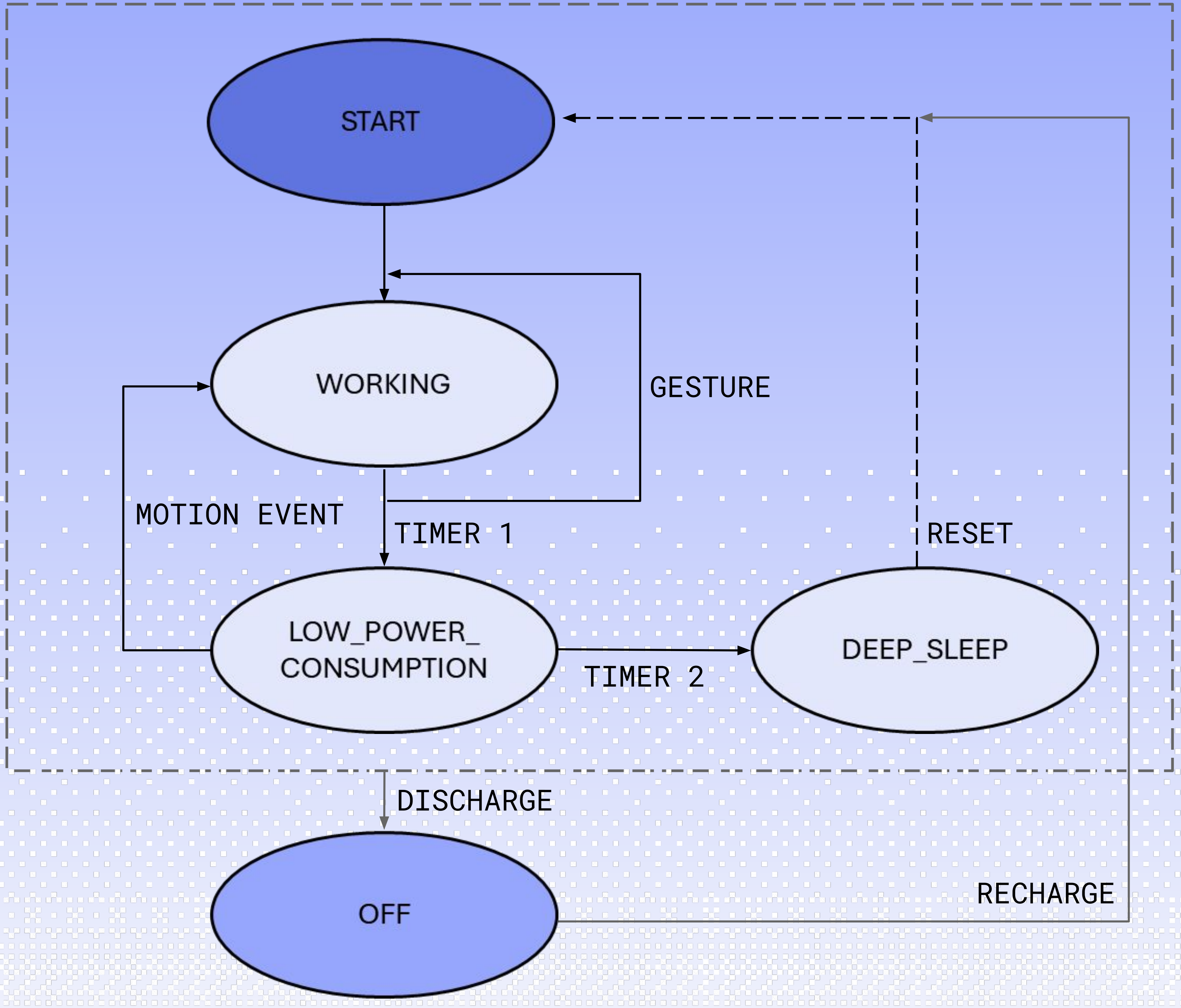


LEGEND

- **GESTURE**: movement or touch detection
- **MOTION EVENT**: movement detection
- **TIMER 1**: 5 min inactivity timeout
- **TIMER 2**: 10 min inactivity timeout

STATES

- **START**: initialization and tasks creation
- **WORKING**: operative
- **LOW_POWER_CONSUMPTION** and **DEEP SLEEP**: power saving
- **OFF**: device powered off

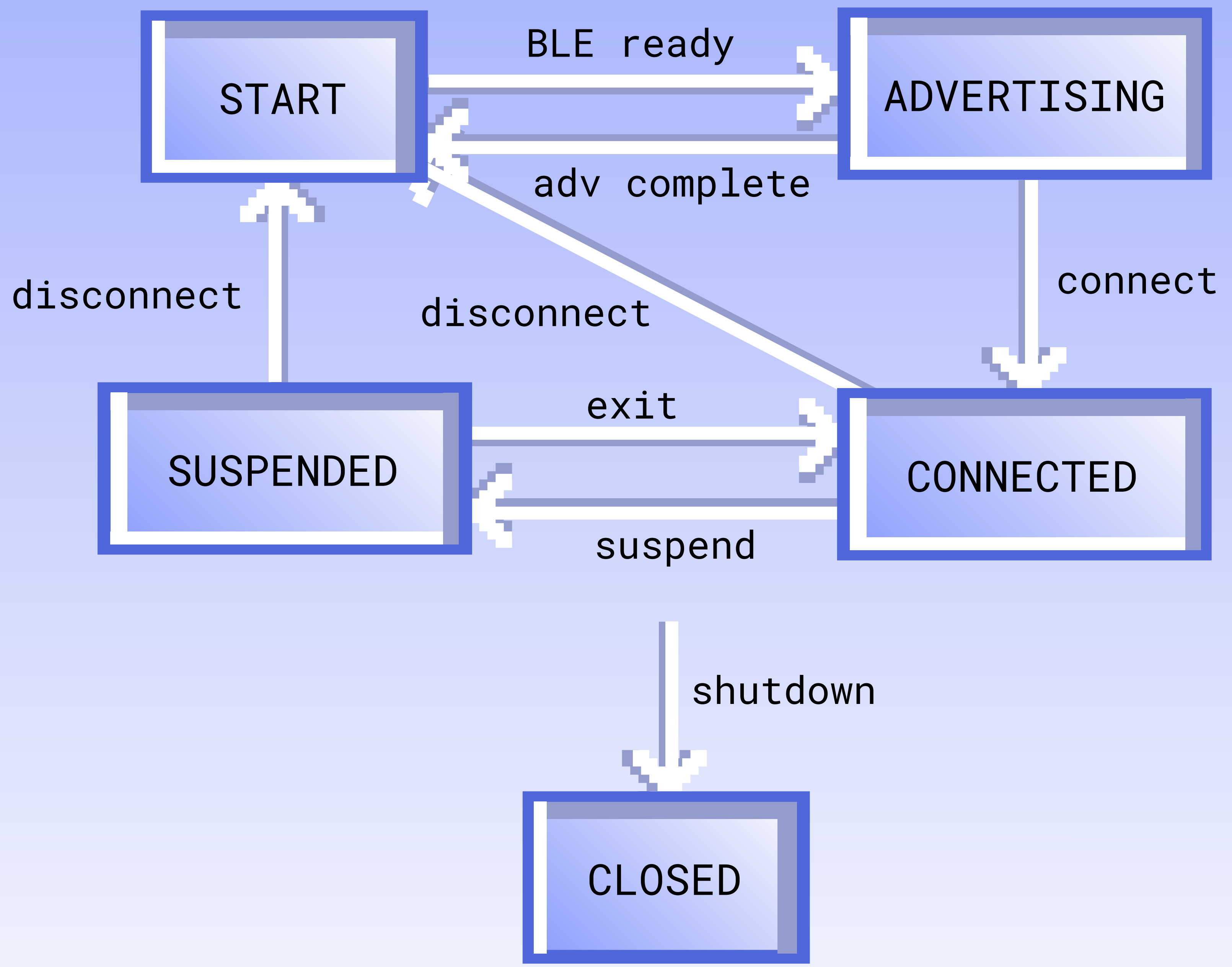
CORE FUNCTIONS

- enter_low_power_mode()
- check_inactivity()
- enter_deep_sleep()

- **GAP connection** (NimBLE)
- **HID over GATT Profile**
  Wrapper around USB HID 1.11 protocol
- **Thread-safe API**
  Control queue
  Data queue
- **Logic composition of messages**
  Saves transmission energy

HID | Battery | Device Info

Connection FSM



Three-Layer Driver Architecture

1. LOW-LEVEL: SPI communication (2MHz)
2. MID-LEVEL: SROM firmware, registers configuration, data reading
3. HIGH-LEVEL: Simplified API

- SPI at 2MHz
- 8-bit delta output

Initialization Process

1. HW RESET
2. GPIO CONFIG
3. SPI SETUP
4. SW RESET
5. PRODUCT ID (0X47)
6. SROM LOAD (ID: 0XE8)

Key Features

Interrupt-Driven (GPIO 18)

Noise Filtering (moving average filter)

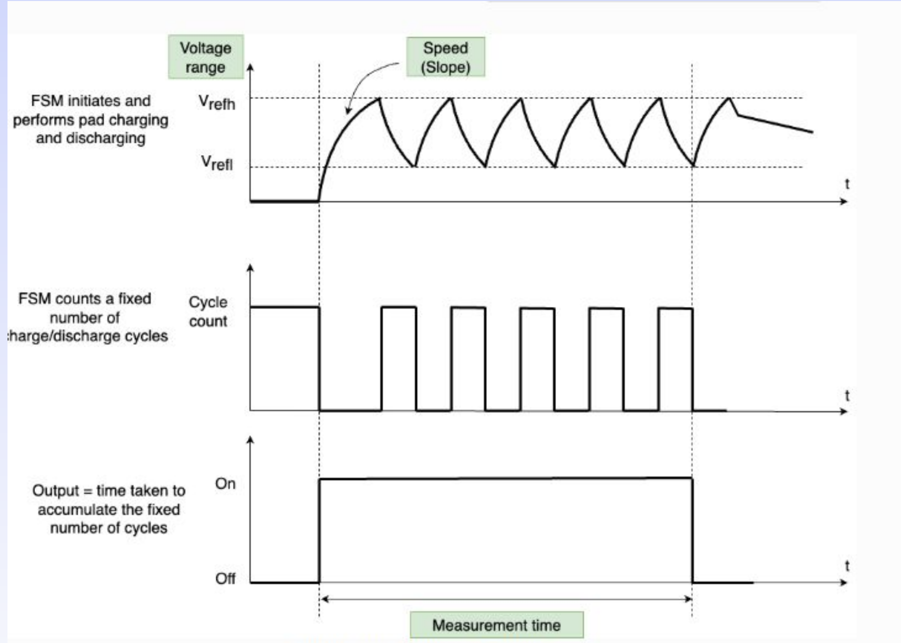
Rich data Output (Δx-Δy, surface quality, lift detection)

GESTURE RECOGNITION PIPELINE

X

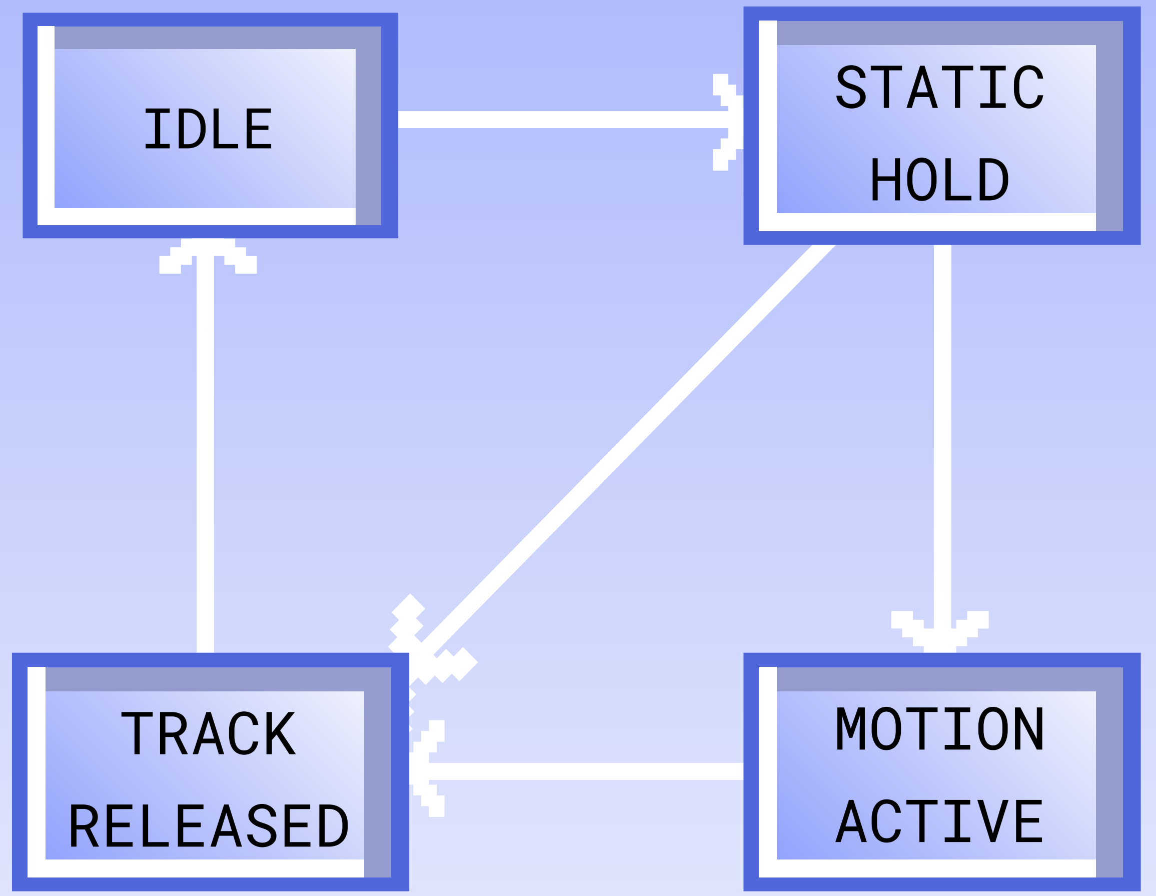
RAW DATA → FILTERING → OVERSAMPLING → BLOB DETECTION → TRACKING → GESTURE RECOGNITION

- **Signal Acquisition & Pre-processing**
  EMWA Filtering and bilinear oversampling
- **Spatial Analysis & Clustering**
  Flood-Fill, centroid mass calculation and finger rejection filter
- **Temporal Tracking & Association**
  Euclidean matching, Tracker persistence logic
- **Final Gesture Recognition**



Touch Sensor Working Principle

Tracker FSM



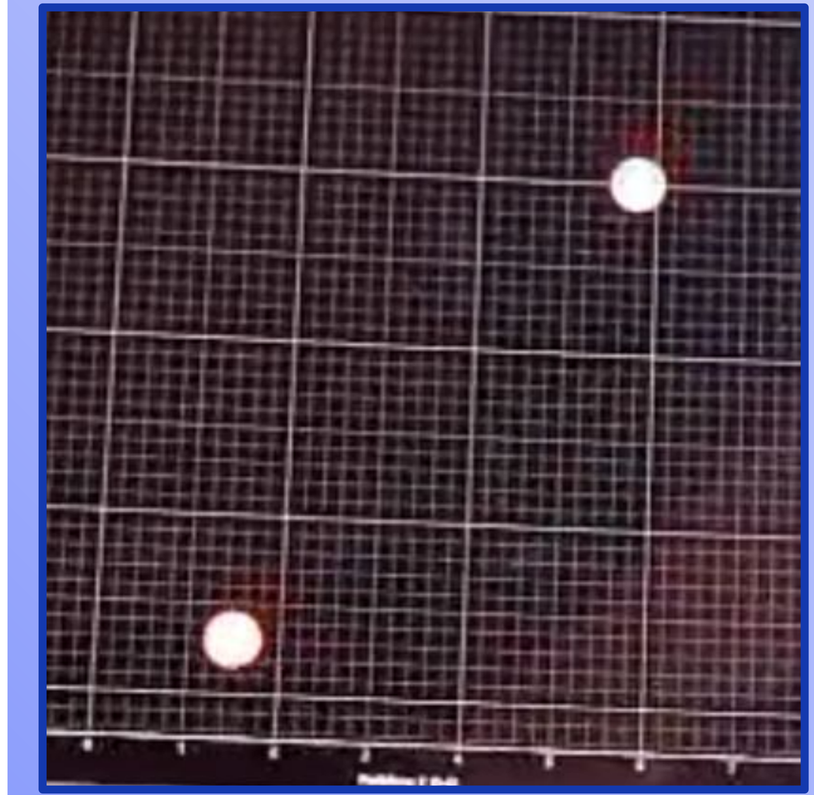
TESTING & FUTURE

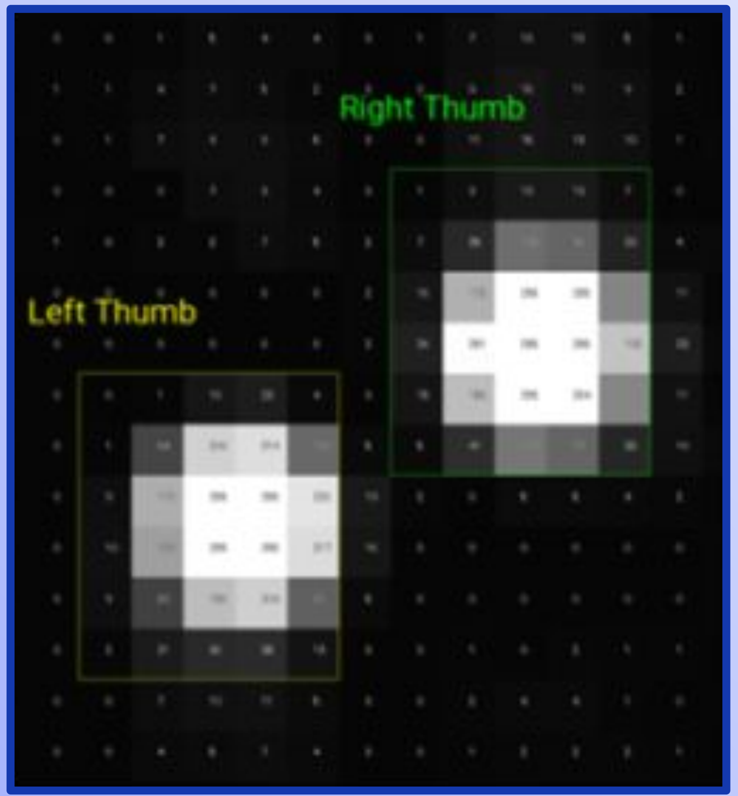
X

TESTING

- Hardware dependent code
- Finite state machines
- Touch gestures pipeline
- Hardware independent code

```
Test 1 - empty queue: Success
Test 2 - cursor motion: Success
Test 3 - mouse buttons: Success
Test 4 - scroll motion: Success
Test 5 - battery level: Success
Test 6 - mixed types & state persistence: Success
```





FUTURE IMPROVEMENTS

- Support Windows and MacOS
- Fix transmission frequency negotiation
- Fix some scroll gesture glitches
- Haptic motor for touch feedback
- Small single PCB for everything
- Flexible PCB for touch matrix
- Improve optical sensor processing
- Support new touch gestures

CONTRIBUTIONS

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ELENA CARMAGNANI

Contribute to: Workflow State Machine design and implementation, low power consumption modes
Code lines: 429

DANIELE MAZZON

Contribute to: Bluetooth HID implementation, battery monitoring system
Code lines: 2272

FEDERICO G. RUBINO

Contribute to: PCB/circuit design and printing, touch gesture recognition pipeline
Code lines: 1256

ILARIA BASANISI

Contribute to: develop the PMW3389 optical sensor driver and the 3D-printed mouse enclosure
Code lines: 1570

THANK YOU FOR THE ATTENTION!

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FAB LAB UNITRENTO

Thanks to Fablab for
the support